

$C_{\emptyset} \text{ as } N^*$

A Minimal Network-Dynamics Model of Phenomenal Consciousness

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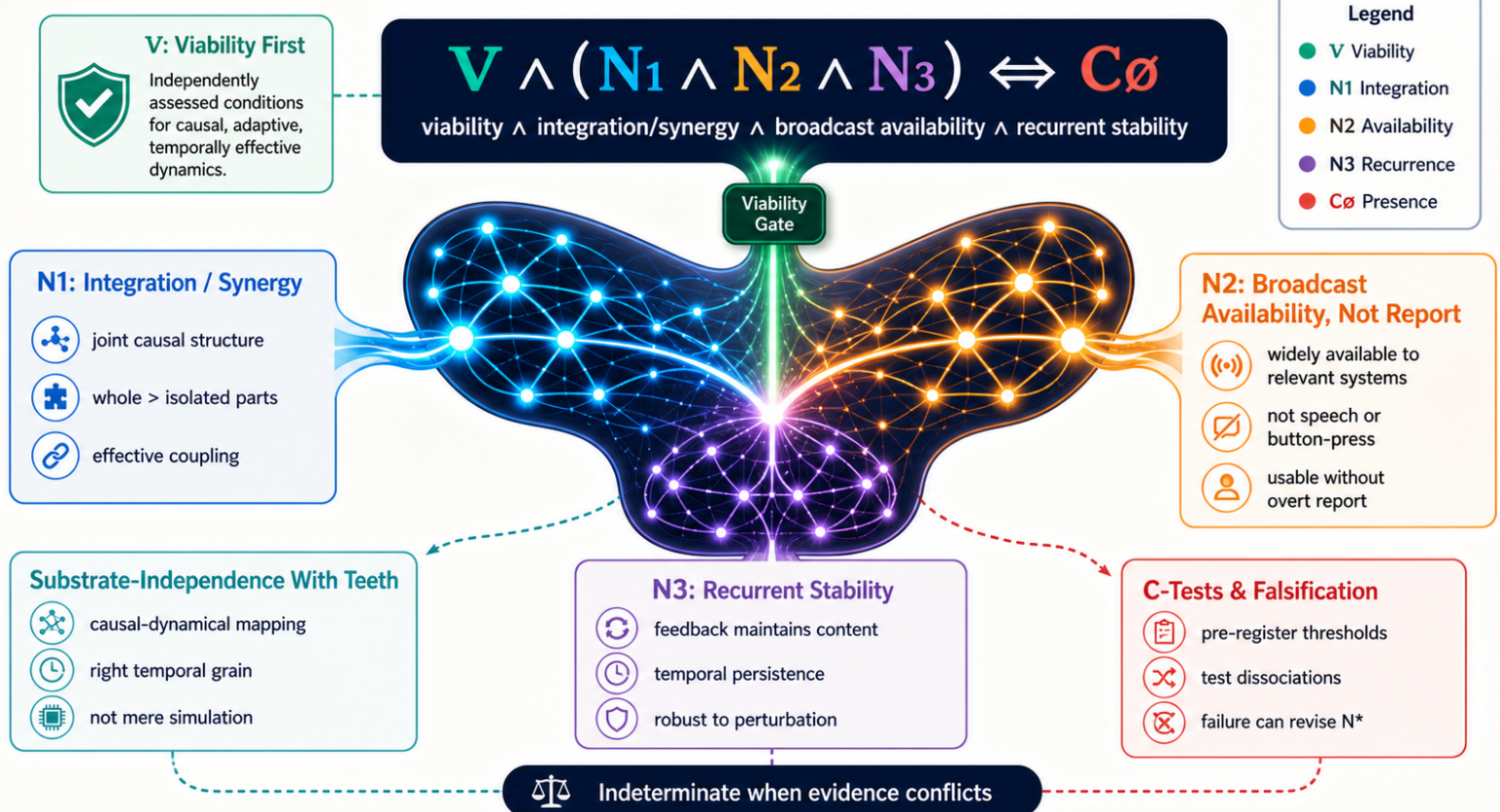
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A visual preview:

$C_{\emptyset} \text{ as } N^*$: Minimal Phenomenal Presence



Abstract

This paper develops an expanded version of the N^* model of phenomenal consciousness. The target is minimal phenomenal presence, denoted $C_\emptyset$: the condition in which there is something it is like for a system to be in a state, without requiring self-reflection, overt report, introspective judgment, or metacognitive access. The central hypothesis is a deliberately risky biconditional: under an independently assessed viability condition $\mathcal{V}$, $C_\emptyset$ obtains if and only if a system instantiates a minimal network-dynamics bundle, formally represented as $N^* = N_1 \wedge N_2 \wedge N_3$. The three conjuncts are integration/synergy (N_1), broadcast availability (N_2), and recurrent stability (N_3). The paper supplies the formal notation, plain-language annotations, a conceptual transition guide for readers arriving from rival frameworks, measurement policy, surrogate-conflict rules, edge-case predictions, a landscape map across major consciousness theories, and objection-handling needed to make the model accessible without weakening its empirical commitments. The proposal is substrate-independent only in a constrained sense: a non-biological or alternative biological substrate qualifies only if it realizes the relevant causal dynamics, at the relevant temporal grain, within pre-registered tolerance bands. The model is not presented as a completed solution to the hard problem or as a full theory of phenomenal quality. It is proposed as a candidate theory of phenomenal presence and as a candidate C-test framework whose value lies in being clear enough to test, constrained enough to fail, and modular enough to be revised by adversarial evidence.

Keywords: phenomenal consciousness; integration; global broadcast; recurrence; network neuroscience; C-tests; substrate independence

$C_\emptyset$ as N^* : A Minimal Network-Dynamics Model of Phenomenal Consciousness

Author Note

This manuscript is a substantially revised and expanded version of the earlier paper titled “ $C_\emptyset$ as N^* : A Minimal Graph-Dynamical Account of Phenomenal Consciousness.” The present version expands the formal annotations, measurement protocol, objection handling, literature positioning, worked cases, preregistration guidance, and appendices. This version supersedes the earlier Academia.edu version for citation and substantive evaluation.

1. Introduction: Why Another Model?

The science of consciousness is not short on theories. Higher-order theories, global workspace theories, recurrent processing theories, predictive processing approaches, and integrated information theory each identify important constraints on conscious states (Seth & Bayne, 2022). The difficulty is that these theories often emphasize different explananda, different neural signatures, and different standards of evidence. Some aim to explain reportable access. Some aim to explain intrinsic phenomenal structure. Some focus on recurrent cortical processing. Others focus on global availability, integrated information, or multiscale dynamics.

The N^* model begins from a modest thought: perhaps a theory of minimal phenomenal presence should not choose only one of these families of constraints. Perhaps phenomenal presence requires a specific joint profile. It must be unified enough not to be a mere heap of local events; available enough not to remain an isolated local fluctuation; and stable enough not to vanish before it can constitute a temporally present state.

This is the guiding idea:

$$N^* := N_1 \wedge N_2 \wedge N_3.$$

Plain-language reading. N^* is the name of the minimal bundle. A system satisfies N^* only when it satisfies all three component conditions: integration/synergy, broadcast availability, and recurrent stability.

Why this matters. The model does not claim that any one condition is novel by itself. Its distinctive claim is conjunctive: none of the three is sufficient alone, and their viable conjunction is the minimal target for $C_\emptyset$.

The central hypothesis is:

$$H_{C_\emptyset} : \mathcal{V} \wedge (N_1 \wedge N_2 \wedge N_3) \iff C_\emptyset. \quad (1)$$

Plain-language reading. A system is phenomenally conscious in the minimal sense exactly when it is viable and instantiates the N^* bundle at the relevant causal and temporal grain.

Why this matters. Equation 1 is intentionally strong. It asserts both sufficiency and necessity. It therefore gives critics two ways to defeat the model: find viable N^* without $C_\emptyset$, or find $C_\emptyset$ without viable N^* .

The model is not offered as an established result. It is offered as a testable conjecture. This distinction matters. A model in consciousness science can be legitimate before it is confirmed if it makes disciplined commitments: what it predicts, what it excludes, how it handles measurement disagreement, and how it would change under evidence. Recent adversarial work in consciousness science has made this demand sharper: theories need quantitative, pre-registered predictions that can be compared directly rather than merely interpreted after the fact (Cogitate Consortium et al., 2025; Doerig et al., 2021).

2. Scope: What $C_\emptyset$ Means and Does Not Mean

By $C_\emptyset$ I mean minimal phenomenal presence: the presence of subjective “what-it-is-like” for a system at a time (Nagel, 1974). This is not the same as self-consciousness, introspection, reportability, attention, intelligence, or wakefulness.

The following distinctions are central:

$$C_\emptyset \neq R, \quad C_\emptyset \neq A, \quad C_\emptyset \neq C_1.$$

Here R denotes overt report, A denotes attention or task-directed access, and C_1 denotes self-reflective or metacognitive consciousness.

Plain-language reading. A subject may be phenomenally conscious without being able to say so, attend to the state, or represent the state as its own state.

Why this matters. This blocks a common distracting assumption: that a model of consciousness must be a model of verbal report or explicit introspection. The present model targets the lower-level presence of experience, not the richer ability to describe or evaluate that experience.

This distinction follows the long-standing contrast between phenomenal and access consciousness (Block, 1995). It also fits methodological concerns from no-report paradigms. If report is required in the experimental design, neural activity associated with planning, memory, decision, or motor output can be mistaken for the neural basis of the experience itself (Tsuchiya et al., 2015). The N^* model therefore treats report as evidence, not as a constituent:

$$R \text{ may indicate } C_\emptyset, \quad \text{but} \quad R \not\equiv C_\emptyset.$$

Hidden assumption being blocked. The model rejects the assumption that the only scientifically useful evidence for consciousness is overt report. It also rejects the opposite overcorrection: that reports are useless. Reports remain valuable evidence in ordinary adults, but they are not part of the definition of $C_\emptyset$.

The model also does not claim to explain every aspect of phenomenology. It is a model of phenomenal presence: whether experience is present at all. It is not yet a complete theory of phenomenal character: why this experience feels like red, pain, spatial depth, anxiety, or melody. Section 16 returns to this limitation and sketches a quality-space extension.

3. A Gentle Logic Primer

Because the model is stated symbolically, the logic should not remain hidden behind the notation. The relevant ideas are simple.

A necessary condition is something that must be present for the target to obtain. Oxygen is necessary for ordinary fire. Without oxygen, no ordinary fire.

A sufficient condition is something that guarantees the target. A lit match in the right conditions is not merely associated with ignition; it can be sufficient for it.

A conjunction is an “and” claim:

$$N_1 \wedge N_2 \wedge N_3.$$

Plain-language reading. N_1 , N_2 , and N_3 must all hold. If any one fails, N^* fails.

A biconditional is an “if and only if” claim:

$$X \iff Y.$$

Plain-language reading. X implies Y , and Y implies X . The claim runs in both directions.

For this model, the two directions are:

$$\mathcal{V} \wedge N^* \implies C_\emptyset, \tag{2}$$

$$C_\emptyset \implies \mathcal{V} \wedge N^*. \tag{3}$$

Equation 2 is the sufficiency direction. It says that if a system is viable and satisfies the N^* bundle, then the model predicts $C_\emptyset$.

Equation 3 is the necessity direction. It says that if $C_\emptyset$ is present, then the model predicts that viable N^* dynamics should be recoverable at the right level of measurement.

What would count against it. The sufficiency direction is threatened by $\mathcal{V} \wedge N^* \wedge \neg C_\emptyset$. The necessity direction is threatened by $C_\emptyset \wedge \neg(\mathcal{V} \wedge N^*)$.

The viability condition $\mathcal{V}$ is included because graph dynamics do not occur in a vacuum. A system can have a formal organization on paper but lack the energy, arousal, coupling, or noise constraints needed to realize the relevant dynamics. The danger is that $\mathcal{V}$ could become a rescue clause. Section 10 therefore defines a rule: $\mathcal{V}$ must be assessed independently before consciousness classification.

Interlude: Conceptual Transition Guide for Rival Frameworks

Readers rarely enter consciousness science as blank slates. A reader trained on IIT may expect a theory to begin from intrinsic phenomenal structure. A GNWT reader may look for ignition, access, and report. A higher-order theorist may expect conscious states to require a

representation of the state as one's own. A recurrent-processing theorist may expect local feedback to be enough. A predictive-processing reader may look for hierarchical inference and prediction-error control. A philosopher shaped by the hard problem may ask whether any network-dynamics account can touch phenomenal feel at all (Chalmers, 1995). Recent theory-comparison work emphasizes that disagreement in the field often begins at this level: theories may appear to compete while partly targeting different explananda, different mechanisms, or different standards of evidence (Mudrik et al., 2025; Seth & Bayne, 2022).

The N^* model should therefore be read with a translation rule:

Do not ask first which familiar theory it resembles. Ask what role each condition plays.

Plain-language reading. The model is not asking the reader to abandon their home framework. It is asking the reader to separate the target, the mechanism, the evidence, and the metaphysical interpretation before judging the biconditional.

The most common conceptual stumbling blocks are these:

- **From IIT.** The tempting assumption is that integration, if properly formalized, should already be enough. N^* accepts integration as indispensable but demotes it from a complete theory to one conjunct. The question becomes whether integration without availability and recurrent stability is phenomenal presence or only structured causal unity.
- **From GNWT.** The tempting assumption is that availability is access, and access is report. N^* accepts system-wide availability but denies that verbal, motor, or executive report is constitutive of $C_\emptyset$. Availability is a network-dynamics role, not a speech act.
- **From recurrent processing.** The tempting assumption is that local feedback is the core of experience. N^* accepts recurrence as the temporal condition for presence but asks whether a stable local loop can be conscious if it remains unintegrated or unavailable to the relevant system.
- **From higher-order and metacognitive theories.** The tempting assumption is that a state is conscious only when the system represents itself as being in that state (Lau & Rosenthal, 2011).

N^* treats that capacity as a candidate condition for richer self-consciousness, C_1 , not for minimal phenomenal presence, C_0 .

- **From predictive processing and free-energy approaches.** The tempting assumption is that the right currency is prediction, precision, or error minimization. N^* can be implemented in predictive-processing terms, but it does not identify consciousness with inference as such. It asks which inferential states are integrated, available, and recurrently stable.
- **From attention-schema or self-model approaches.** The tempting assumption is that subjective awareness depends on an internal model of attention or self-related control (Graziano & Webb, 2015). N^* can treat attention schemas as possible contributors to N_2 or C_1 , but it does not require that every C_0 state include an explicit attention model.
- **From illusionism.** The tempting assumption is that the real scientific target is not phenomenality but the system's disposition to represent or report itself as phenomenally conscious (Frankish, 2016). N^* does not settle that philosophical dispute. It remains realist about C_0 as its target, while allowing illusionist-style explanations to be tested as accounts of reports about C_0 .
- **From embodied, affective, or evolutionary approaches.** The tempting assumption is that consciousness is inseparable from organismic regulation, action, affect, or learning history (Ginsburg & Jablonka, 2019). N^* does not dismiss these. It places their minimal enabling role in $\mathcal{V}$ and their content-shaping role in the graph dynamics, while leaving open whether some forms of embodiment are necessary for particular kinds of C_0 .
- **From panpsychist or strongly functionalist approaches.** The tempting assumption is that the right abstract organization may be enough, perhaps very widely instantiated. N^* resists this by requiring causal realization, temporal grain, effective coupling, and independently assessed viability.

Why this matters. Many apparent objections are really category shifts. A reader may reject N^* because it does not explain report, self-consciousness, intrinsic quality, organismic life, or the

metaphysical ground of experience. Those may be legitimate demands on a complete theory, but the present target is narrower: minimal phenomenal presence under explicit empirical risk.

The working discipline is therefore:

1. Identify the target: $C_\emptyset$, not report, self-consciousness, intelligence, or full qualitative structure.
2. Identify the role: N_1 supplies unity, N_2 supplies availability, N_3 supplies temporal stability, and $\mathcal{V}$ supplies independent dynamical viability.
3. Identify the risk: each role must have empirical surrogates, conflict rules, and failure cases.
4. Identify the remaining question: if a reader's preferred framework requires more than this, the added requirement should be stated as an additional conjunct rather than smuggled into the definition of $C_\emptyset$.

4. Network-Dynamics Background

Let:

$$G = (V, E)$$

be a causal graph. The elements of V are nodes, and the elements of E are directed or weighted causal relations among them. In a biological brain, nodes may be neurons, populations, cortical regions, thalamic nuclei, brainstem systems, or other functionally defined units, depending on the scale of analysis. Edges are not merely anatomical wires. They represent effective interactions: which parts make a difference to which other parts, with what direction, strength, and timing.

This distinction matters:

- **Structural connectivity** concerns relatively stable physical pathways.
- **Functional connectivity** concerns statistical dependence among activities.
- **Effective connectivity** concerns directed causal influence.

The N^* model is primarily about graph dynamics. It asks not merely whether a system has the right static shape, but whether the right causal interactions occur over the right time window.

4.1 Temporal Grain

Consciousness is not normally tested at an instantaneous mathematical point. The model therefore treats “at a time” as a windowed claim. A content c is evaluated over a time window Δ long enough to test integration, availability, and recurrence. In many perceptual paradigms, the relevant window may lie roughly in the range of 100–600 ms, though the appropriate value will depend on modality, species, task, and measurement technique.

Hidden assumption being blocked. The model blocks the assumption that consciousness must be either a timeless property or an arbitrarily instantaneous event. It is a temporally extended dynamical condition.

4.2 Whole-Brain Domain

The causal graph G is not restricted to cortex. Subcortical structures, including thalamic systems, brainstem arousal systems, and basal forebrain systems, may be elements of V . Edges in E include cortico-subcortical interactions. The hub set H in N_2 and loop set L in N_3 may include or even be dominated by subcortical nodes, depending on species, state, and task. This follows the multiscale direction of recent consciousness theory, which increasingly treats conscious states as involving recurrent, integrative, and representational processes across levels rather than as the product of one privileged cortical site (Storm et al., 2024).

5. The Model in One View

The model has two layers: the bundle and the hypothesis.

First, the bundle:

$$N^* := N_1 \wedge N_2 \wedge N_3. \quad (4)$$

Second, the hypothesis, restated from Equation 1, is:

$$H_{C_\emptyset} : \mathcal{V} \wedge N^* \iff C_\emptyset.$$

Plain-language reading. Minimal phenomenal consciousness is present exactly when a viable system satisfies all three N^* conditions.

Why this matters. The hypothesis is a compact way to say something precise: consciousness is not identified with integration alone, broadcast alone, recurrence alone, report, behavior, or biological material. It is identified with a viable pattern of integrated, available, recurrent graph dynamics.

What would count against it. Any well-controlled case of viable N^* without $C_\emptyset$ counts against sufficiency. Any well-controlled case of $C_\emptyset$ without viable N^* counts against necessity.

The model uses “minimal” in a specific sense. It does not mean trivial, simple, or easy to detect. It means that the three conjuncts are proposed as the fewest conditions needed for phenomenal presence. Adding metacognition, language, self-modeling, autobiographical memory, or explicit judgment may explain richer conscious capacities, but these are not included in $C_\emptyset$ unless evidence shows that $C_\emptyset$ itself cannot occur without them.

6. N_1 : Integration/Synergy

N_1 is the unity condition. A conscious content is not merely a set of independent local signals. It is bound enough to be a content for the system.

Formally:

$$\exists U \subseteq V \text{ such that } \text{Synergy}_\Delta(U) \geq \tau_1. \quad (5)$$

Plain-language reading. Within the relevant time window, some subset of the system must bind information together in a way that exceeds a pre-registered threshold.

Why this matters. Without integration, the model predicts fragments rather than unified phenomenal presence. A distributed set of independent signals may carry information, but it does not yet constitute a unified conscious field or content.

What would count against it. If subjects retain unified conscious contents while all credible integration/synergy measures collapse below threshold, N_1 is too strong. If high N_1 routinely appears in systems independently classified as nonconscious while N_2 and N_3 are absent, the model can survive; if high N_1 plus viable N_2 and N_3 appears without $C_\emptyset$, the sufficiency direction is threatened.

The intuitive role of N_1 overlaps with the integration emphasis in integrated information theory (Albantakis et al., 2023; Tononi, 2008). The difference is that N^* does not treat integration

alone as sufficient. It treats integration as one necessary condition inside a broader dynamical package.

6.1 What Kind of Integration?

The model is intentionally pluralistic at the level of estimators but not permissive at the level of decision rules. Candidate measures include perturbational complexity, multivariate synergy, O-information class measures, and model-based effective connectivity that captures higher-order dependence beyond pairwise correlation (Casali et al., 2013). The shared target is not “activity” in general, but a pattern in which the parts jointly carry information that is not reducible to their independent contributions.

One useful mechanistic gloss is this: increasing $\text{Synergy}_\Delta(U)$ raises the minimum causal cut needed to disrupt content c . If a local perturbation can destroy the content by disrupting one small component, the content is fragile. If the content is integrated across a causally interacting subset, it becomes more resistant to local noise and fragmentation.

6.2 Why Integration Alone Is Not Enough

A reader sympathetic to IIT may wonder why N_2 and N_3 are needed. The answer is that a highly integrated process can remain local, unavailable, or dynamically unstable. The N^* model therefore makes a stricter claim:

$$N_1 \not\Rightarrow C_\emptyset.$$

Plain-language reading. Integration alone does not imply phenomenal presence.

This does not deny that integration matters. It denies that integration, by itself, is the whole story. In the model, integration supplies unity; it does not by itself supply availability or temporal presence.

7. N_2 : Broadcast Availability Without Reportism

N_2 is the availability condition. A conscious content must become available across enough of the system quickly enough to play a system-level role. But this is not the same as overt report.

Formally:

$$\exists H \subseteq V \text{ such that } \min_{v \in H} \text{PC}(v) \geq \theta, \quad (6)$$

$$\frac{|\text{Reach}_\Delta(c; H)|}{|V|} \geq \alpha \quad \text{and} \quad \text{latency}_\Delta(c; H) \leq \lambda. \quad (7)$$

Plain-language reading. There must be a hub or backbone structure through which a content becomes broadly available within the relevant time window.

Why this matters. A locally integrated content that never becomes available to the wider system remains isolated. The model predicts that such a content is not yet a full instance of $C_\emptyset$, though it may be a proto-content, unconscious fragment, or indeterminate case.

What would count against it. If $C_\emptyset$ clearly persists when a content fails all credible measures of system-wide availability, N_2 is too strong. If report is lost while no-report markers and N^* remain intact, the model predicts that $C_\emptyset$ may remain despite absent report.

This condition draws from global workspace and global neuronal workspace theories, which emphasize ignition, availability, and the sharing of information across specialized processors (Dehaene & Changeux, 2011; Mashour et al., 2020). The N^* model differs in two respects. First, it does not require prefrontal dominance as a fixed anatomical commitment. Second, it does not identify availability with report.

7.1 Availability, Access, and Report

The model distinguishes three notions:

- **Availability:** a content is causally poised to influence multiple systems.
- **Access:** a content is usable by task, memory, control, or decision systems.
- **Report:** a content is expressed through behavior, speech, button press, or another output channel.

The relation is evidential, not definitional:

$$R \Rightarrow \text{evidence for availability,} \quad \text{but} \quad R \not\equiv N_2.$$

Hidden assumption being blocked. This blocks the assumption that N_2 smuggles in access consciousness. N_2 is a network-dynamics availability condition. Reports can help detect it, but they are not what makes it true.

7.2 Posterior and Subcortical Possibilities

The model does not assume that H must always be a prefrontal hub set. The Cogitate adversarial collaboration found results that challenged simple versions of both GNWT and IIT, including evidence relevant to timing, localization, and posterior content information (Cogitate Consortium et al., 2025). The N^* model should therefore allow H to vary by content, modality, and species. For some visual contents, a posterior-dominant backbone may satisfy N_2 ; for other states, thalamic or cortico-subcortical routing may be essential.

The model’s commitment is not anatomical chauvinism. It is the inequality in Equation 7: broad-enough reach, fast-enough latency, and appropriate hub participation for the content under test.

8. N_3 : Recurrent Stability

N_3 is the temporal presence condition. A conscious content must not be merely a feedforward spark. It must persist in a recurrent or metastable regime long enough to become present for the system.

Formally:

$$\exists L \subseteq V \text{ such that } t_{\min} \leq \text{dwell}(c; L) \leq t_{\max}. \quad (8)$$

Plain-language reading. There must be recurrent loops supporting the content for a bounded interval. The content must last long enough to be present, but it need not and should not persist indefinitely.

Why this matters. Without recurrence, the model predicts transient processing without phenomenal presence. Without boundedness, the model risks confusing consciousness with pathological fixation, seizure-like persistence, or inert memory.

What would count against it. If conscious contents persist under conditions where feedback and recurrent dwell are genuinely absent, N_3 is too strong. If recurrent dwell alone appears without

integration and availability, the model predicts no full $C_\emptyset$.

This condition overlaps with recurrent processing theory (Lamme, 2006). The N^* model adopts recurrence as necessary but not sufficient:

$$N_3 \not\Rightarrow C_\emptyset.$$

8.1 Why Dwell Is Bounded

The lower bound $t_{\min}$ prevents the model from counting momentary causal flickers as phenomenal presence. The upper bound $t_{\max}$ prevents the model from treating indefinite persistence as automatically better. Conscious contents are typically metastable: they stabilize, influence the system, and then update or decay. In this sense, phenomenal presence is neither instantaneous nor frozen.

Operationally, $\text{dwell}(c; L)$ may be estimated as the median survival time before the autocorrelation of c 's trajectory over L drops below a pre-registered fraction f , or as the plateau duration in a fitted metastable hidden Markov model.

9. Why the Three Conditions Must Be Joint

The model's main argumentative burden lies here. Integration, availability, and recurrence each solve a different problem. Remove any one, and a familiar failure mode appears.

Plain-language reading. The model does not ask whether a system has one impressive property. It asks whether it has the right conjunction of properties.

This conjunctive strategy is partly motivated by the current theoretical landscape. If the literature contains plausible evidence for integration, broadcast, and recurrence, a minimal model should ask whether these are rival explanations or mutually constraining dimensions (Seth & Bayne, 2022; Storm et al., 2024). The N^* model takes the second route.

10. Viability $\mathcal{V}$ and Why It Is Not a Rescue Clause

The viability condition is:

$\mathcal{V}$ = independently assessed arousal/energy/noise/dynamical-capacity condition.

Plain-language reading. The system must be capable of sustaining the dynamics under test.

Why this matters. Without $\mathcal{V}$, the model could overgeneralize to inert diagrams, dead tissue, disconnected simulations, or mathematical structures that instantiate no real dynamics.

What would count against it. If $\mathcal{V}$ is adjusted after observing whether a system appears conscious, it becomes a rescue clause and weakens the model.

The rule is:

$\mathcal{V}$ must be assessed before $C_\emptyset$ classification.

Possible operational markers include:

- arousal-system integrity;
- metabolic and energetic viability;
- noise bounds compatible with sustained dynamics;
- perturbational responsiveness;
- capacity for recurrent stabilization;
- species-appropriate or substrate-appropriate dynamical support.

The C-test literature is especially relevant here because many important cases involve systems that cannot provide ordinary adult human reports: infants, fetal systems, nonhuman animals, disorders of consciousness, organoids, and artificial systems (Bayne et al., 2024). The N^* model treats $\mathcal{V}$ as part of the test architecture, not as an after-the-fact explanation for failed predictions.

11. Measurement Policy and Surrogate Conflict

Each N_i is defined abstractly, but empirical testing requires surrogate measures. This creates a danger: if many measures are available, one might choose whichever favors the model. The expanded model therefore needs a conflict rule.

For component N_i , let $M_{i,1}, M_{i,2}, \dots, M_{i,k}$ be the pre-registered measures for that component. Then:

$$\text{if } \text{Var}(M_{i,1}, \dots, M_{i,k}) > \kappa_i, \quad N_i = \text{Indeterminate.} \quad (9)$$

Plain-language reading. If the measures for a component disagree too much, the model does not get to cherry-pick the favorable measure. It returns an indeterminate result.

Why this matters. This makes N^* behave like a real empirical model rather than a flexible interpretive vocabulary.

The decision pipeline is:

1. Pre-register the paradigm, temporal window, system grain, and candidate surrogates.
2. Estimate $\mathcal{V}$ independently.
3. Estimate N_1 , N_2 , and N_3 using held-out or cross-validated thresholds.
4. Apply the surrogate-conflict rule in Equation 9.
5. Classify the case as $C_\emptyset$ predicted, $C_\emptyset$ not predicted, or indeterminate.

12. Pre-registration and Parameter Robustness

The parameters of the model are not meant to be tuned after the outcome is known. They are meant to be pre-registered, estimated with held-out data, and subjected to sensitivity analysis.

The main parameters are:

$$(\tau_1, \theta, \alpha, \lambda, t_{\min}, t_{\max}, f, \kappa_1, \kappa_2, \kappa_3).$$

Why this matters. The model's credibility depends less on the initial numerical values than on whether those values are pre-registered, justified, and robust. If tiny parameter shifts reverse the conclusion, the model is brittle.

13. Falsification Handles

The model makes three kinds of empirical commitments: sufficiency, necessity, and exclusion.

13.1 Sufficiency Failure

The sufficiency direction fails if:

$$\mathcal{V} \wedge N^* \wedge \neg C_\emptyset. \quad (10)$$

Plain-language reading. A viable system satisfies all three N^* conditions, yet there is strong reason to deny phenomenal presence.

13.2 Necessity Failure

The necessity direction fails if:

$$C_\emptyset \wedge \neg(\mathcal{V} \wedge N^*). \quad (11)$$

Plain-language reading. Phenomenal consciousness is present, yet the viable N^* bundle is absent at the relevant grain.

13.3 Exclusion Events

If N^* holds within the pre-registered tolerance bands, then none of the following should occur:

1. PCI/complexity collapse for the target content and window;
2. absence of late recurrent markers;
3. failure of cross-module decoding of c ;
4. surrogate conflict beyond κ_i while still claiming a positive result.

Observation of an exclusion event under intact N^* threatens the sufficiency direction.

Observation of $C_\emptyset$ under persistent exclusion events threatens the necessity direction.

14. Edge Cases and Adversarial Predictions

The model should make predictions before the outcome is known. Table 5 gives a first-pass map. The entries are intentionally cautious: in some cases, the model predicts indeterminacy rather than a confident yes or no.

14.1 Adversarial Tasks

The following task families would be especially useful:

1. **Offset vs. onset signatures.** Test whether late offset signatures dissociate from prefrontal ignition while N_1 and N_3 remain posteriorly intact. The model permits preserved $C_\emptyset$ if N^* is satisfied by posterior or non-prefrontal hubs.
2. **Posterior-only broadcast.** In visual paradigms, allow H to be posterior-dominant. The model predicts sufficiency if reach and latency constraints pass with posterior hubs.
3. **No-report awareness.** Use binocular rivalry, eye-tracking, optokinetic responses, or other no-report markers to test whether N^* tracks conscious content independently of overt report (Tsuchiya et al., 2015).
4. **Feedback disruption.** Disrupt recurrent loops while preserving early feedforward processing. The model predicts loss or degradation of $C_\emptyset$ when N_3 falls below threshold.
5. **Perturbational complexity under anesthesia.** Test whether decreases in N_1 and N_3 jointly predict loss of $C_\emptyset$ better than either measure alone (Casali et al., 2013).

15. How N^* Fits Across the Consciousness Landscape

The N^* model should not be framed as if it floats above the literature. Its components are deliberately drawn from live theoretical traditions, but the resulting conjunction changes the role each component plays. The current landscape includes mechanistic neural theories, metacognitive theories, representational and quality-space theories, embodied or evolutionary approaches, C-test frameworks, and deflationary or metaphysical positions. A useful way to situate N^* is to ask whether a neighboring view supplies a component, an implementation, a rival target, or a philosophical interpretation.

Recent reviews stress that the major theories do not always disagree about a single shared object. They may differ over whether the explanandum is phenomenal presence, reportable access,

phenomenal structure, self-awareness, cognitive control, or the introspective impression of phenomenality itself (Mudrik et al., 2025; Seth & Bayne, 2022). The N^* model tries to reduce that ambiguity by declaring its target first: $C_\emptyset$, minimal phenomenal presence.

15.1 Integrated Information Theory

IIT aims to account for phenomenal existence and phenomenal structure in physical and mathematical terms (Albantakis et al., 2023). The N^* model shares IIT's insistence that unity and integration matter. It differs by refusing to treat integration alone as sufficient for $C_\emptyset$. In N^* , integration supplies unity; it does not by itself supply broadcast availability or recurrent temporal presence.

15.2 Global Neuronal Workspace Theory

GNWT emphasizes ignition, amplification, and global availability (Mashour et al., 2020). The N^* model adopts availability as N_2 , but it weakens two tempting overreadings. First, N_2 is not equated with report. Second, N_2 is not assumed to require one fixed anatomical locus across all tasks and species. The relevant question is whether the content meets reach and latency constraints through an appropriate hub/backbone set.

15.3 Recurrent Processing

Recurrent processing theories emphasize feedback and local recurrent loops (Lamme, 2006). The N^* model treats recurrence as necessary for temporal presence but not sufficient for consciousness. A stable loop that remains isolated and unintegrated is not enough.

15.4 Predictive Processing and Free Energy

Predictive processing and free-energy approaches emphasize hierarchical inference, prediction error, and self-organizing dynamics (Friston, 2010). The N^* model can be interpreted through these mechanisms, especially when recurrent loops stabilize shared priors and broadcast availability coordinates inference across scales. But it does not identify $C_\emptyset$ with prediction error minimization as such.

15.5 Higher-Order and Metacognitive Views

Higher-order approaches emphasize a system's capacity to represent its own mental states (Lau & Rosenthal, 2011). The N^* model does not deny the importance of metacognition. It locates metacognition at a richer level, C_1 , rather than building it into the minimal conditions for C_0 . A metacognitive system may be conscious in richer ways, but the question here is whether minimal phenomenal presence requires that added structure.

15.6 Attention Schema and Self-Model Views

Attention-schema and self-model approaches explain subjective awareness partly through a system's internal model of its own attention, control, or self-related processing (Graziano & Webb, 2015). The N^* model can incorporate such models as contributors to availability, control, or metacognitive enrichment. It does not make them mandatory for C_0 . If future evidence shows that minimal phenomenal presence requires an attention schema, the model should add that as an explicit conjunct rather than quietly treating it as already contained in N_2 .

15.7 C-tests

Bayne et al. emphasize the need for validated tests for consciousness across populations that cannot be assessed by ordinary adult report (Bayne et al., 2024). The N^* model can be read as a candidate C-test framework. Its promise is that it gives a structured way to combine markers; its risk is that those markers may fail to validate against difficult cases. That risk is a virtue if it is made explicit.

15.8 Quality-Space and Representational Approaches

Quality-space approaches emphasize the relational structure of phenomenal character: red feels more like orange than like green or pain because experiences occupy structured similarity spaces (Fleming & Shea, 2024). This is crucial for extending N^* beyond presence. Section 16 treats quality-space structure as a possible next layer rather than as part of the minimal C_0 condition.

15.9 Embodied, Affective, and Evolutionary Approaches

Embodied and affective approaches stress that consciousness is not merely a detached cortical calculation. It is often tied to organismic regulation, action-readiness, interoception, affect, and learning. Evolutionary approaches such as unlimited associative learning treat certain flexible learning capacities as markers for the emergence of consciousness (Ginsburg & Jablonka, 2019). The N^* model can accommodate much of this by treating the conscious system as a viable, situated dynamical system rather than as a disembodied graph. The unresolved question is whether embodiment is only an enabling condition for many biological systems, or whether some organismic form of embodiment is necessary for C_0 in any substrate.

15.10 Illusionism and Deflationary Views

Illusionist approaches argue that phenomenal consciousness, as commonly conceived, is an introspective misrepresentation, and that the scientific task is to explain why systems judge or represent themselves as having phenomenal properties (Frankish, 2016). The N^* model is not illusionist: it takes C_0 as a real target. But it can still learn from illusionism by separating phenomenal presence from claims about phenomenal presence. A report that “I am conscious” is evidence to be explained; it is not identical to C_0 .

15.11 Panpsychist, Fundamentalist, and Universalist Views

Panpsychist or fundamentalist views treat consciousness, proto-consciousness, or phenomenal properties as very widely distributed or metaphysically basic. Universalist projects seek general constraints that could apply across substrates (Kleiner & Tull, 2024). The N^* model is more restrictive. It does not deny that consciousness might be substrate-independent, but it refuses to infer consciousness from mere existence, mere complexity, or mere abstract structure. It requires viable, causally realized integration, availability, and recurrence at the relevant grain.

15.12 Summary of the Fit

Conceptually, N^* is best understood as a mid-level bridge theory. It is not a metaphysical theory of why experience exists at all, not a report theory, not a metacognitive theory, and not a

complete quality theory. It is a testable model of the minimal dynamical profile under which phenomenal presence is predicted. It borrows from integration theories, workspace theories, recurrence theories, and multiscale dynamical theories, then makes those borrowings accountable through a biconditional, a viability condition, and explicit failure modes.

IIT supplies unity pressure;

GNWT supplies availability pressure;

RPT supplies temporal pressure.

N^* turns those pressures into a conjunctive empirical claim.

16. Phenomenal Quality: The Deliberate Limitation

The strongest remaining weakness of the N^* model is that it targets phenomenal presence rather than phenomenal character. It says when experience is present, not yet why the experience has the specific character it does.

This limitation should be stated plainly:

N^* models presence; N^* does not yet fully model quality.

Plain-language reading. The model asks whether there is something it is like. A further account is needed to explain what it is like.

A possible extension is:

$$Q(c) = F(\mathcal{G}_c, \mathcal{S}_c, \mathcal{R}_c). \quad (12)$$

Here $Q(c)$ denotes the qualitative character of content c , $\mathcal{G}_c$ denotes content-specific graph geometry, $\mathcal{S}_c$ denotes similarity or quality-space relations, and $\mathcal{R}_c$ denotes recurrent stabilization profile.

Plain-language reading. The character of an experience may depend not only on whether N^* is present, but on where the content sits in a structured space of discriminable similarities and differences.

Why this matters. This allows the paper to avoid overclaiming. A theory of presence can be valuable even if it is not yet a complete theory of color, pain, emotion, or spatial phenomenology.

17. Substrate Independence With Teeth

The model is substrate-independent, but not in the loose sense that any abstractly similar diagram is conscious. Let a second system realize a causal graph $\tilde{G}$, and let φ be a mapping from $\tilde{G}$ to G . The mapping condition is:

$$\text{dist}(G, \varphi(\tilde{G})) \leq \epsilon. \quad (13)$$

This is not enough by itself. The system must also preserve the relevant dynamics:

$$(\tau_1, \theta, \alpha, \lambda, t_{\min}, t_{\max}, f) \quad \text{within pre-registered tolerance bands.}$$

Plain-language reading. Same abstract shape is not enough. The alternative substrate must realize the right causal dynamics at the right grain.

Why this matters. This blocks both neural chauvinism and empty functionalism. The model does not require carbon-based neurons, but neither does it grant consciousness to any system with a superficially similar graph.

The mapping must preserve:

- directionality;
- effective coupling;
- temporal grain;
- nontrivial causal closure;
- dynamic parameter realization;
- substrate-appropriate viability $\mathcal{V}$.

Non-biological systems, organoids, analog neuromorphic devices, or quantum simulators qualify only if they meet the mapping and dynamical criteria. The model therefore avoids panpsychist proliferation by refusing purely syntactic isomorphism.

18. Potential Objections and Replies

Objection 1: This is just IIT plus GNWT plus recurrence.

The model does borrow constraints from these traditions. That is not a defect unless the borrowing is unconstrained. The distinctive claim is the biconditional: viable $N_1 \wedge N_2 \wedge N_3$ is jointly sufficient and individually necessary for $C_\emptyset$. The model is therefore not just a list of themes. It is a conjunction with exclusion rules, measurement policies, and falsification handles.

Objection 2: N_2 smuggles in access consciousness.

N_2 is not reportability. It is a network-dynamics availability condition. Reports can help detect it, but no-report paradigms are central to testing it (Tsuchiya et al., 2015). The model predicts $C_\emptyset$ from $\mathcal{V} \wedge N^*$, not from verbal or motor output.

Objection 3: The thresholds are arbitrary.

The initial thresholds are provisional. Their scientific role is to be pre-registered, tested, and revised. A model with explicit thresholds is more vulnerable than a verbal theory with hidden thresholds. Parameter brittleness counts against the model.

Objection 4: $\mathcal{V}$ makes the model unfalsifiable.

It would, if $\mathcal{V}$ were assigned after the fact. The model forbids that. $\mathcal{V}$ must be assessed independently before $C_\emptyset$ classification.

Objection 5: The model ignores qualia.

It does not ignore phenomenal quality; it brackets it. The model targets phenomenal presence first. Section 16 identifies quality-space structure as the next layer needed for a fuller theory.

Objection 6: The model overgeneralizes to AI or simple systems.

The model does not infer consciousness from computational sophistication, linguistic behavior, or graph isomorphism alone. It requires substrate-appropriate viability, effective coupling, temporal grain, and the dynamic realization of N_1 , N_2 , and N_3 .

Objection 7: We cannot know $C_\emptyset$ in non-reporting systems.

Correct, if “know” means direct certainty. But science often proceeds with structured tests under uncertainty. The model should therefore output not only positive and negative predictions, but also indeterminate classifications when evidence is insufficient or surrogate measures conflict.

Objection 8: The explanandum is still too thin.

A critic may say that $C_\emptyset$ is so minimal that the model wins only by lowering the bar. The reply is that thinness is a methodological choice, not an evasion. The paper separates phenomenal presence from report, metacognition, and detailed quality because those are often conflated. The model becomes stronger, not weaker, if later evidence shows that one of those richer capacities must be added as a fourth conjunct.

Objection 9: The biconditional is too strong for early science.

This is partly correct. The biconditional is not presented as established doctrine. It is a deliberately exposed conjecture. A weaker model could claim only that N^* is correlated with consciousness, but that would be less informative. The biconditional earns its keep by making both directions vulnerable: viable N^* without $C_\emptyset$ and $C_\emptyset$ without viable N^* are both serious threats.

Objection 10: Consciousness may be graded rather than binary.

The model can accommodate graded evidence without abandoning the biconditional. The binary form applies to a specified thresholded classification within a paradigm; the underlying measures can vary continuously. In difficult cases, the output should be graded or indeterminate until the paradigm justifies a threshold. A future version could replace the sharp biconditional with a probabilistic version:

$$P(C_\emptyset \mid \mathcal{V}, N_1, N_2, N_3) \geq p^*.$$

That would be a revision of the decision rule, not a rejection of the conjunctive architecture.

Objection 11: The model mistakes neural correlates for constitutive conditions.

This is a serious risk for any empirical theory of consciousness. N^* tries to reduce the risk by defining its components causally rather than merely correlationally. Structural association is not

enough; the relevant graph must realize effective integration, availability, and recurrent stability. Perturbation, intervention, and adversarial dissociation tests are therefore more important than passive decoding alone.

Objection 12: Third-person graph measures cannot capture first-person experience.

The model does not claim that graph measures are themselves first-person data. It claims that they may identify the causal-dynamical conditions under which first-person presence obtains. This leaves the hard problem open in the strong metaphysical sense (Chalmers, 1995). The model's narrower ambition is to be a scientifically useful bridge between first-person evidence, behavioral evidence, and dynamical mechanisms.

Objection 13: The model inherits all the weaknesses of its source theories.

It inherits some risks, but not all commitments. From IIT it inherits pressure toward integration, not IIT's full mathematical metaphysics. From GNWT it inherits availability, not reportism or fixed prefrontal localization. From recurrent theories it inherits temporal stabilization, not the claim that recurrence alone is sufficient. The model should be rejected if the conjunction fails; it should not be rejected merely because one source theory overclaimed.

Objection 14: The model may underplay affect, embodiment, and life.

This is a legitimate pressure point. The current model treats viability $\mathcal{V}$ as the place where arousal, energy, regulatory capacity, and substrate support enter. But affective and embodied theorists may argue that such factors are not merely enabling conditions; they may partly constitute $C_\emptyset$. The model should therefore test whether organismic regulation and interoceptive-affective loops are optional contributors, necessary components of $\mathcal{V}$, or candidates for an additional conjunct.

Objection 15: Indeterminate outputs are ethically unsatisfying.

They are unsatisfying, but they are sometimes the only honest scientific output. In infants, disorders of consciousness, animals, organoids, or artificial systems, premature confidence can be morally dangerous in both directions. The model's indeterminate category should therefore be paired with a precautionary ethical policy: uncertainty in classification should not be treated as

evidence of absence when welfare-relevant decisions are at stake.

19. Open Questions

The model is strongest if it advertises its vulnerabilities. Key open questions include:

- **Surrogate validity.** Which measures best track the intended components of N^* ?
- **Ground truth.** How should the model validate against cases where report is unavailable or unreliable?
- **Subcortical dominance.** Can subcortical hubs and recurrent loops satisfy N_2 and N_3 in states with minimal cortical involvement?
- **Quality mapping.** Can quality-space geometry be linked to content-specific graph dynamics?
- **Parameter portability.** Which parameters transfer across modalities, species, and substrates?
- **Boundary and grain.** How should the model identify the boundaries of a conscious system in split, nested, or distributed cases?
- **Framework translation.** Which objections survive once report, access, metacognition, phenomenal quality, and minimal presence are kept apart?
- **Embodiment and affect.** Are organismic regulation and affective valence enabling conditions inside $\mathcal{V}$, or do they require an additional conjunct?
- **Ethical indeterminacy.** What precautionary policy should follow when the model cannot confidently classify a welfare-relevant system?

20. Conclusion

The N^* model proposes a precise, testable biconditional for minimal phenomenal consciousness:

$$H_{C_\emptyset} : \quad \mathcal{V} \wedge (N_1 \wedge N_2 \wedge N_3) \iff C_\emptyset.$$

Its claim is modest in target but strong in form. It does not aim to explain self-reflection, verbal report, intelligence, or the full quality of experience. It aims to identify the minimal network-dynamics conditions under which phenomenal presence obtains.

The model's value lies in three features. First, it is integrative: it treats integration, availability, and recurrence as mutually constraining rather than as isolated theoretical slogans. Second, it is annotated: every symbolic claim has an accessible interpretation and an empirical risk. Third, it is falsifiable: viable N^* without $C_\emptyset$, or $C_\emptyset$ without viable N^* , would force revision or rejection.

The path forward is therefore empirical. Pre-register the thresholds. Validate the surrogates. Use no-report and report-minimized paradigms. Test disorders of consciousness, dreaming, anesthesia, rivalry, split systems, and non-biological substrates only under disciplined mapping criteria. The model should be taken seriously exactly to the degree that it remains exposed to failure.

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Table 1

Core notation in the expanded model.

Symbol	Formal meaning	Accessible gloss
$G = (V, E)$	Causal graph	The system's interacting parts and their causal relations.
X_t	System state at time t	What the system is doing at a moment.
U	Integrating subset	The parts binding a content together.
H	Hub/backbone set	Nodes that make a content broadly available.
L	Recurrent-loop set	Feedback loops that stabilize a content.
c	Decodable content	A representational pattern under test.
Δ	Time window	The interval over which the dynamics are evaluated.
$\mathcal{V}$	Viability condition	Arousal, energy, noise, and dynamical capacity required for testing.

Table 2

Predicted failure modes when one or more conjuncts are absent.

Present	Missing	Model prediction
$N_1 + N_2$	N_3	Integrated and available content fails to stabilize as phenomenal presence.
$N_1 + N_3$	N_2	Stable local unity remains isolated from global availability.
$N_2 + N_3$	N_1	Available and persistent activity lacks unified content.
N_1 only	N_2, N_3	Local integration without full phenomenal presence.
N_2 only	N_1, N_3	Broadcast without coherent, temporally present content.
N_3 only	N_1, N_2	Persistence without conscious unity or availability.

Table 3
Measurement families and conflict handling.

Component	Candidate measures	Conflict rule
N_1	PCI-style perturbational complexity, multivariate synergy, O-information, effective connectivity beyond pairwise terms	If integration measures diverge beyond κ_1 , classify N_1 as indeterminate.
N_2	Participation coefficient, rich-club measures, time-resolved decoding, reach/latency across modules	If broadcast measures diverge beyond κ_2 , classify N_2 as indeterminate.
N_3	Recurrent effective connectivity, dwell time, metastable HMM plateaus, late feedback markers	If recurrence measures diverge beyond κ_3 , classify N_3 as indeterminate.

Table 4

Illustrative parameter bands and robustness checks. These are not universal constants; they are to be locked per paradigm.

Parameter	Illustrative prior band	Estimator/check
τ_1	80th–95th percentile of baseline; expected decrease under deep anesthesia	PID/O-information; PCI-style perturbation
θ	0.45–0.65	Participation coefficient or related hub measure
α	0.35–0.60 of nodes or modules in Δ	Time-resolved decoding and reach
λ	100–300 ms visual; 200–500 ms cross-modal	Time-to-criterion
$t_{\min}, t_{\max}$	150–600 ms window	Survival estimates; HMM plateaus
f	0.2–0.4	Autocorrelation decay
κ_i	Paradigm-specific tolerance	Surrogate-disagreement threshold

Table 5

Edge-case predictions for the N^ model.*

Case	Likely profile	Model prediction	Pressure point
Normal waking perception	$\mathcal{V}+$, N_1+ , N_2+ , N_3+	C_0 present	Baseline calibration case.
REM dreaming	$\mathcal{V}+$, N_1+ , N_2 partial/+, N_3+	C_0 present if N_2 meets dream-state availability criteria	Report absent until waking.
Deep NREM sleep	$\mathcal{V}$ partial, N^* low or sparse	C_0 absent, sparse, or indeterminate	Tests minimal phenomenal experience claims.
General anesthesia	$\mathcal{V}$ low/partial, N_1 , N_2 , N_3 reduced	C_0 absent or sharply reduced	PCI and perturbation tests.
Locked-in syndrome	$\mathcal{V}+$, N^*+	C_0 present	Report not required.
Vegetative state	variable $\mathcal{V}$ and N^*	absent or indeterminate unless N^* recoverable	C-test validation.
Minimally conscious state	fluctuating $\mathcal{V}$ and N^*	graded or episodic C_0 predictions	State instability.
Blindsight	$\mathcal{V}+$, visual N^* incomplete for blind field	no visual C_0 for blind-field content	Access vs presence.
Binocular rivalry	$\mathcal{V}+$, N^* tracks dominant content	C_0 tracks dominant percept	No-report paradigms.
Split brain	$\mathcal{V}+$, lateralized N^* profiles	possible partitioned or partially independent C_0	Boundary and grain.
Organoids	$\mathcal{V}$ uncertain, N^* unknown	no verdict without $\mathcal{V}$ and N^* tests	Avoids overclaiming.
AI systems	substrate-specific $\mathcal{V}$ uncertain	no verdict without causal-dynamic mapping	Avoids syntactic panpsychism.

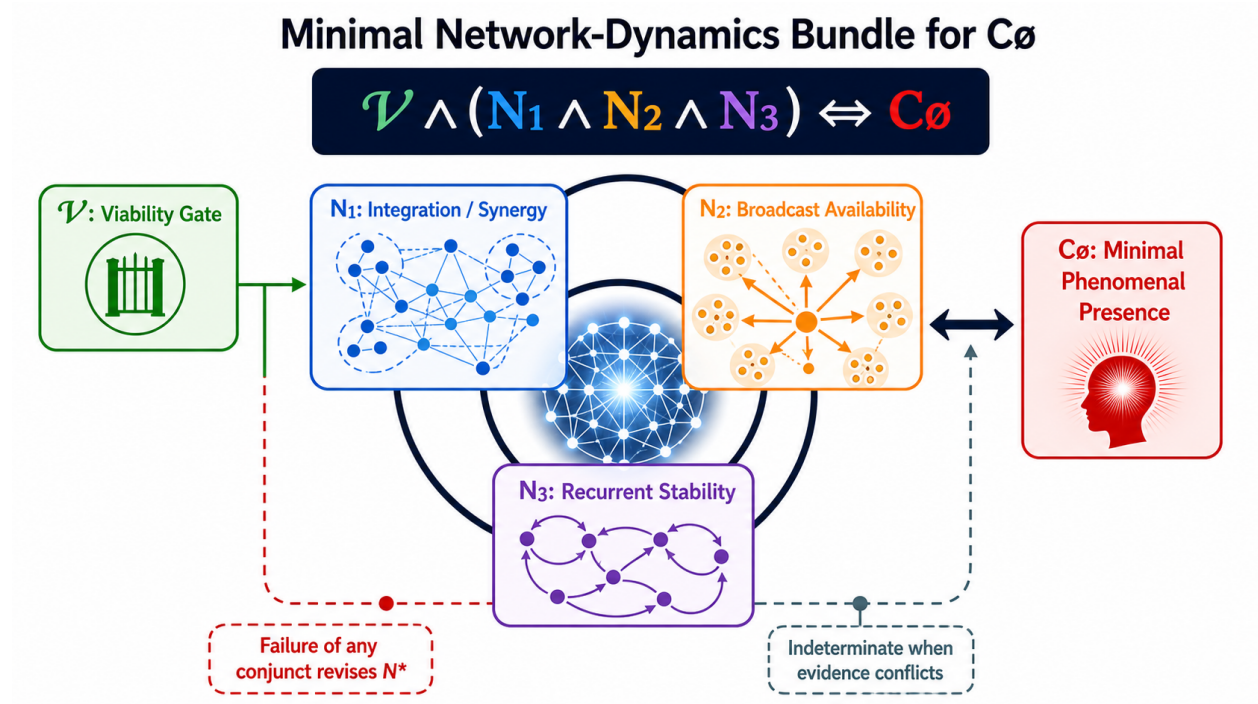


Figure 1

Formal schematic of the expanded N^* model. The independently assessed viability condition $\mathcal{V}$ gates the three conjuncts of the minimal network-dynamics bundle: integration/synergy (N_1), broadcast availability (N_2), and recurrent stability (N_3). Their viable conjunction is offered as a testable biconditional target for C_0 , with conflicting surrogate evidence treated as indeterminate and failures treated as reasons to revise the model.

Appendix A

Glossary and Symbol Key

This appendix is meant to remove a common source of friction: the same word can carry different meanings in philosophy, cognitive science, neuroscience, artificial intelligence, and graph theory. The model is easiest to read when its terms are treated as roles inside a testable account rather than as inherited slogans from any one framework. The entries below therefore give both the technical role and the intended plain-language reading.

Table A1

Core symbols and their intended readings.

Symbol	Technical role	Plain-language reading
$C_\emptyset$	Minimal phenomenal presence	There is something it is like for the system, without requiring self-report, introspection, or a self-concept.
C_1	Enriched or reflective consciousness	A richer level involving metacognition, self-reference, report, or ownership judgments.
N^*	Minimal network-dynamics bundle	The conjunction of integration, availability, and recurrent stability.
N_1	Integration/synergy	The content is unified by interactions that are not reducible to isolated parts.
N_2	Broadcast availability	The content is made available across a relevant system-level backbone, not merely trapped in a local pocket.
N_3	Recurrent stability	The content persists through feedback long enough to count as temporally present.

Symbol	Technical role	Plain-language reading
$\mathcal{V}$	Viability condition	The system has the arousal, energy, coupling, noise profile, and dynamical capacity needed for the test to be meaningful.
$G = (V, E)$	Causal graph	The system's interacting parts and the effective relations among them.
U	Integrating subset	The parts that jointly bind or unify a content.
H	Hub/backbone set	Nodes or modules through which a content becomes broadly available.
L	Recurrent-loop set	Feedback circuits that stabilize a content over a time window.
c	Candidate content	The percept, affective tone, image, memory, or state being evaluated.
Δ	Evaluation window	The time span over which the model asks whether the relevant dynamics hold.
R	Report	Speech, button press, or other behavior used as evidence, not as the definition of consciousness.
A	Attention/access marker	A marker of selection or task availability that may correlate with $C_\emptyset$ without being identical to it.
τ_1, θ, α	Component thresholds	Pre-registered cutoffs for integration, hub participation, and reach.
$\lambda, t_{\min}, t_{\max}, f$	Temporal parameters	Latency, lower and upper dwell limits, and decay criteria for recurrent stability.
κ_i	Conflict tolerance	The maximum allowed disagreement among surrogate measures before a component becomes indeterminate.

Symbol	Technical role	Plain-language reading
φ	Cross-substrate mapping	A proposed relation between two systems' causal roles, temporal profiles, and effective dynamics.

Table A2

Conceptual distinctions that should not be collapsed.

Term	Do not collapse it into	Use in the present model
Phenomenal presence	Report, access, intelligence, wakefulness	The minimal target: whether experience is present at all.
Phenomenal character	Mere presence	The specific way an experience feels; treated as a further quality-space layer.
Access	Verbal report	Functional availability to other processes; report is only one possible expression.
Viability	A rescue clause	A prior constraint on whether the system can realize the relevant dynamics.
Substrate independence	Anything with a similar diagram	The same causal-dynamical role at the right grain, not merely a syntactic resemblance.
Indeterminacy	Failure or evasion	A disciplined result when evidence is insufficient or surrogate measures conflict.
Surrogate measure	The target itself	An operational proxy for a component of N^* , to be validated and stress-tested.
Consciousness landscape	A menu of labels	A set of neighboring theories that may differ in target, mechanism, evidence, or metaphysics.

Appendix B

Logic Walkthrough of the Biconditional

The central hypothesis can look more intimidating than it is:

$$H_{C_\emptyset} : \mathcal{V} \wedge (N_1 \wedge N_2 \wedge N_3) \iff C_\emptyset.$$

It says that, under the model, minimal phenomenal presence obtains exactly when a viable system instantiates the three-part N^* bundle at the relevant grain. This is not a stipulative definition of $C_\emptyset$. It is a risky modeling claim about the conditions under which $C_\emptyset$ is predicted.

A biconditional contains two directional commitments:

$$\mathcal{V} \wedge N^* \implies C_\emptyset$$

and

$$C_\emptyset \implies \mathcal{V} \wedge N^*.$$

The first is the sufficiency direction. If a viable system satisfies the bundle, the model predicts $C_\emptyset$. The second is the necessity direction. If $C_\emptyset$ is genuinely present, the model predicts that viable N^* dynamics should be recoverable at the correct level of description.

Table B1

How to read possible outcomes under the biconditional.

Pattern	Model status	Why it matters
$\mathcal{V}+$ and N^*+ with evidence of $C_\emptyset$	Expected success	The bundle tracks the target in the case under study.
$\mathcal{V}+$ and N^*+ with strong evidence against $C_\emptyset$	Sufficiency threat	The left side is present but the right side is absent. The model has overpredicted.

Pattern	Model status	Why it matters
Strong evidence of C_0 with persistent N^* failure	Necessity threat	The right side is present but the left side is absent. The model has made the bundle too demanding.
N_1 only, N_2 only, or N_3 only	Not enough	The model denies that any single impressive property is sufficient.
Failed or unmeasured $\mathcal{V}$	Not licensed or not predicted	If viability is absent, C_0 is not predicted. If viability is unassessed, the test is incomplete.
Major surrogate conflict	Indeterminate	The model should not convert disagreement among measures into false certainty.

Three cautions prevent common misreadings. First, the biconditional is not a claim that present neuroscience can measure every term perfectly. It is a claim about what would have to be true if the model is right. Second, the right-hand side, C_0 , remains the target; it is not reduced to a report or a score. Third, the model can return an indeterminate classification when evidence is poor while still preserving a sharp hypothesis about the underlying condition.

Appendix C

Measurement and Classification Protocol

The protocol below turns the model into a repeatable C-test candidate. Its purpose is to prevent after-the-fact interpretation. The investigator should specify the system, content, time window, measures, thresholds, and conflict rules before looking at the critical outcome.

Table C1

Stepwise classification protocol for applying N^ .*

Stage	Action	Guardrail
1. Bound the system	Specify the candidate system, nodes, edges, content c , and evaluation window Δ .	Do not let the boundary expand after the data are known.
2. Assess $\mathcal{V}$	Evaluate arousal, energy, coupling, noise, sensorimotor or regulatory support, and recording quality.	Assess viability independently of the desired consciousness classification.
3. Estimate N_1	Apply integration or synergy measures selected before analysis.	Require evidence of effective integration, not merely high activity or correlation.
4. Estimate N_2	Test availability across the relevant hub/backbone set H .	Do not equate availability with verbal report or fixed prefrontal localization.
5. Estimate N_3	Test recurrent stability across loop set L within the dwell window.	Distinguish recurrent stabilization from mere persistence of a stimulus trace.
6. Apply conflict rules	Compare surrogate measures against the pre-registered κ_i tolerances.	If measures disagree too strongly, classify the component as indeterminate.
7. Classify	Return $C_\emptyset$ predicted, $C_\emptyset$ not predicted, or indeterminate.	Preserve uncertainty rather than forcing a binary verdict in poor evidence conditions.

Stage	Action	Guardrail
8. Stress-test	Run sensitivity analyses across thresholds, windows, and plausible graph boundaries.	A conclusion that flips under tiny defensible changes should be treated as fragile.

The decision rule can be stated compactly:

Classify $C_\emptyset$ as predicted only if $\mathcal{V}$, N_1 , N_2 , and N_3 all pass without unresolved conflict.

If $\mathcal{V}$ is known to fail, the model does not predict $C_\emptyset$ for that system, content, and window. If $\mathcal{V}$ cannot be assessed, or if any component is indeterminate, the scientifically appropriate output is indeterminate. This is not a retreat from the biconditional; it is a separation between the model's ontological claim and the evidence available in a particular test.

Appendix D

Framework Translation Matrix

Readers often evaluate a new model by silently importing the assumptions of the framework they already trust. The table below makes those imports explicit. The aim is not to flatten rival theories into N^* , but to show where N^* accepts a pressure, resists an inference, or asks for a discriminating test.

Table D1

How N^ translates assumptions from neighboring frameworks.*

Framework	Likely starting assumption	What N^* accepts	What N^* resists or tests
IIT	Integration is central to phenomenal existence.	Unity and irreducible joint structure matter.	Integration alone is not sufficient for C_0 without availability and recurrence.
GNWT	Conscious contents become globally available.	Availability is a necessary network-dynamics role.	Availability is not identical to report, nor fixed to one anatomical site.
Recurrent processing	Feedback is the core mechanism of conscious perception.	Recurrent stabilization is necessary for temporal presence.	Local recurrence alone may remain isolated or unintegrated.
Predictive processing	Conscious states arise within hierarchical inference and precision control.	Prediction and error control can implement the three components.	Inference as such is not identified with C_0 .

Framework	Likely starting assumption	What N^* accepts	What N^* resists or tests
Higher-order theory	Consciousness requires a representation of one's mental state.	Metacognition may enrich consciousness and support C_1 .	Minimal C_0 is not assumed to require self-representation.
Attention-schema or self-model views	Awareness depends on an internal model of attention or self.	Self-models may shape availability, control, and report.	A self-model is not built into the minimal condition unless evidence requires it.
C-test approaches	Non-reporting systems require validated tests.	N^* is a candidate structured test framework.	The test must validate against hard cases rather than merely fit familiar adults.
Quality-space theories	Phenomenal character has structured similarity relations.	Quality structure is a needed next layer.	Presence and character should not be conflated.
Embodied and affective theories	Consciousness is tied to life, regulation, action, and value.	Viability and graph dynamics may include organismic regulation and affective loops.	The model tests whether these are enabling, constitutive, or content-shaping.
Illusionism	The scientific target is the system's judgment that it is conscious.	Reports and self-ascriptions require explanation.	The model remains realist about C_0 as the target.
Panpsychist or universalist views	Consciousness or proto-consciousness may be very widely instantiated.	Substrate independence remains possible in constrained form.	Mere existence, complexity, or abstract isomorphism is not enough.

Appendix E

Worked Case Studies

The following cases are not offered as final diagnoses. They are examples of how the model should discipline its own reasoning. Each case forces the same sequence: bound the system, assess viability, estimate the three components, apply conflict rules, and classify only at the end.

Table E1

Worked examples of N^ classification logic.*

Case	Likely component profile	Model output	Conceptual lesson
Normal waking perception	$\mathcal{V}$ is intact; integration, availability, and recurrence usually pass for attended or dominant contents.	C_0 predicted for the tested content.	The baseline case calibrates measures but should not define the theory by report.
REM dreaming	Viability is altered but often sufficient; integration and recurrence may be strong; availability must be evaluated by dream-state criteria.	C_0 predicted if N_2 passes within the dreaming system.	Report can be delayed until waking without making the dream unconscious.
Deep NREM sleep	Viability and N^* may be sparse, fluctuating, or content-specific.	Absent, sparse, or indeterminate depending on the window.	The model must allow minimal phenomenal experience claims to be tested rather than dismissed.
General anesthesia	Viability, integration, and recurrent complexity are expected to fall under sufficient depth.	C_0 not predicted when the bundle collapses.	Perturbational and recurrent measures should jointly outperform any single marker.

Case	Likely component profile	Model output	Conceptual lesson
Locked-in syndrome	Viability and N^* may remain intact while motor report is blocked.	C_0 predicted.	Report failure is not consciousness failure.
Vegetative or unresponsive wakefulness syndrome	Viability may be partial; N^* evidence may be absent, weak, or covert.	Indeterminate or absent unless recoverable N^* is shown.	The model must be conservative but not dismissive in non-reporting patients.
Minimally conscious state	Viability and N^* may fluctuate across time and content.	Episodic or graded predictions, often with indeterminate windows.	State instability matters; a single negative window should not overgeneralize.
Blindsight	Viability may be intact, but blind-field visual content may lack the full bundle.	No visual C_0 predicted for the blind-field content if N^* fails.	Accurate forced-choice behavior need not imply phenomenal presence.
Binocular rivalry	Viability is intact; N^* should track the dominant percept rather than the suppressed stimulus.	C_0 predicted for the dominant content.	No-report markers can help separate content from response.
Split brain	Viability may be intact; N^* may partition across hemispheric or task-specific boundaries.	Possibly partitioned C_0 profiles.	Boundary and grain decisions become central rather than peripheral.
Organoids	Viability, effective coupling, and temporal grain are uncertain.	No confident verdict without disciplined $\mathcal{V}$ and N^* tests.	The model blocks both overclaiming and easy dismissal.

Case	Likely component profile	Model output	Conceptual lesson
Artificial systems	Abstract architecture may resemble a graph, but viability and causal dynamics must be established.	No verdict without substrate-specific mapping and intervention tests.	Functional similarity must be causal-dynamical, not merely behavioral or syntactic.

Two features of these cases are important. First, the model is content-relative: a system may satisfy N^* for one content or modality and not another. Second, the model is window-relative: a system may pass during one interval and fail during another. This makes the model less rhetorically simple, but more faithful to the unstable cases that motivate C-tests in the first place.

Appendix F

Expanded Objections and Replies

The main text gives concise replies to the most important objections. This appendix states the strongest version of several objections before giving the model's reply. The goal is not to make the objections disappear, but to show which would require clarification, revision, or empirical defeat.

Table F1

Expanded objection map for N.*

Objection	Strongest form	Model reply	Residual pressure
Hybrid without novelty	The model just combines IIT, GNWT, and recurrent processing.	The novelty is the exposed conjunction, viability rule, conflict policy, and biconditional failure structure.	The model must produce predictions its source theories would not jointly force.
Report smuggling	N_2 secretly reintroduces access consciousness and report.	N_2 is availability in graph dynamics; report is only evidence. No-report paradigms are central tests.	Availability measures must be designed so motor planning does not dominate them.
Threshold arbitrariness	Any numerical cutoff will look conventional.	Thresholds are paradigm-locked, cross-validated, and stress-tested rather than universal constants.	If small reasonable shifts change outcomes, the classification is fragile.

Objection	Strongest form	Model reply	Residual pressure
Viability escape hatch	$\mathcal{V}$ can explain away failures.	$\mathcal{V}$ must be assessed independently before classification.	The paper should keep specifying viability criteria for each substrate class.
Qualia ignored	A theory of consciousness that omits phenomenal character misses the point.	The model targets presence first and treats quality as a further layer.	A full theory eventually needs a quality-space extension.
Artificial-system overreach	A graph model will overattribute consciousness to AI systems.	Mere graph isomorphism is rejected; causal realization, temporal grain, and viability are required.	Non-biological viability remains underdeveloped and should be conservative.
Non-report uncertainty	We cannot know $C_\emptyset$ in infants, animals, patients, organoids, or machines.	The model gives structured uncertainty rather than certainty.	Ethical policy must not treat uncertainty as absence.
Thin explanandum	Minimal presence is too weak to satisfy the real explanatory demand.	Thinness is deliberate; conflating presence with report, selfhood, and quality obscures testing.	The model should be presented as a layer, not the whole theory.
Biconditional too strong	Early science should avoid if-and-only-if claims.	The biconditional is a conjecture whose strictness creates falsification routes.	A probabilistic variant may become necessary.

Objection	Strongest form	Model reply	Residual pressure
Graded consciousness	Consciousness may not be binary.	Measures can be continuous while classification uses pre-registered thresholds and indeterminacy.	Borderline cases may require a graded extension.
Correlation vs constitution	Graph measures might track correlates, not conditions.	Intervention and perturbation are prioritized over passive association.	Constitutive interpretation depends on causal evidence.
Hard problem	Third-person dynamics cannot explain why experience exists.	The model is a bridge theory of presence, not a metaphysical solution to the hard problem.	Philosophical dissatisfaction remains legitimate.
Embodiment underplayed	Life, affect, and regulation may be constitutive, not merely enabling.	These factors enter through $\mathcal{V}$ and possibly through graph dynamics.	Future tests may require an affective or organismic conjunct.

Appendix G

Preregistration Template for Testing N^*

This template is designed to make the model easier to test adversarially. A completed preregistration should make clear what would support the model, what would count against it, and what would force an indeterminate result.

Table G1

Fields for a preregistered N^ test.*

Field	Specify before analysis	Reason
Target claim	The system, content c , population, task, and time window Δ .	Prevents shifting the target after the outcome is known.
System boundary	Nodes, modules, sensors, anatomical regions, or computational units included in G .	Boundary choices can determine whether N^* appears to hold.
Viability criteria	Independent criteria for arousal, energy, coupling, noise, and recording quality.	Keeps $\mathcal{V}$ from becoming a rescue clause.
N_1 measures	Integration, synergy, PCI-style, or effective-connectivity estimators.	Makes unity testable rather than impressionistic.
N_2 measures	Hub participation, reach, latency, decoding across modules, or backbone availability.	Separates availability from report and from local activation.
N_3 measures	Recurrent effective connectivity, dwell time, metastable plateaus, or feedback markers.	Tests whether the content is temporally present rather than transient.
Thresholds	Values for $\tau_1, \theta, \alpha, \lambda, t_{\min}, t_{\max}, f$, and κ_i .	Prevents outcome-driven tuning.

Field	Specify before analysis	Reason
Conflict policy	The exact conditions under which measures produce indeterminacy.	Makes disagreement among surrogates informative rather than inconvenient.
Exclusion events	Observations that would count against sufficiency or necessity.	Gives the model real empirical risk.
Control contrasts	Report vs no-report, stimulus-matched, state-matched, and perturbational comparisons.	Helps separate consciousness markers from attention, memory, and motor output.
Sensitivity plan	Parameter sweeps, alternative graph boundaries, and held-out validation.	Identifies brittle conclusions.
Ethical policy	How indeterminate welfare-relevant cases will be handled.	Prevents uncertainty from being misused as evidence of absence.
Revision rule	What model change would follow from each failure pattern.	Keeps failure from being absorbed rhetorically.

A compact preregistration statement can be written as follows:

For system S , content c , and window Δ , we will classify $C_\emptyset$ as predicted only if $\mathcal{V}$, N_1 , N_2 , and N_3 pass the pre-registered thresholds without unresolved surrogate conflict. We will classify $C_\emptyset$ as not predicted if $\mathcal{V}$ is adequate and one or more necessary components fail robustly. We will classify the result as indeterminate if viability is unassessed, if any component measure exceeds its conflict tolerance, or if sensitivity analysis shows that the conclusion depends on a narrow and unjustified parameter choice.